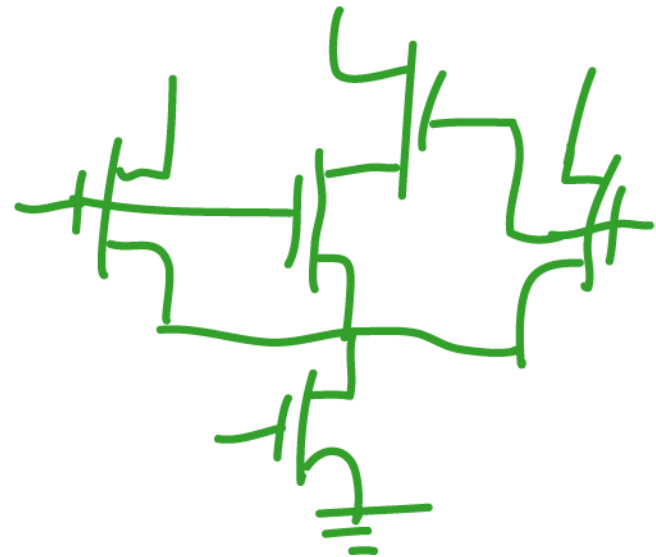
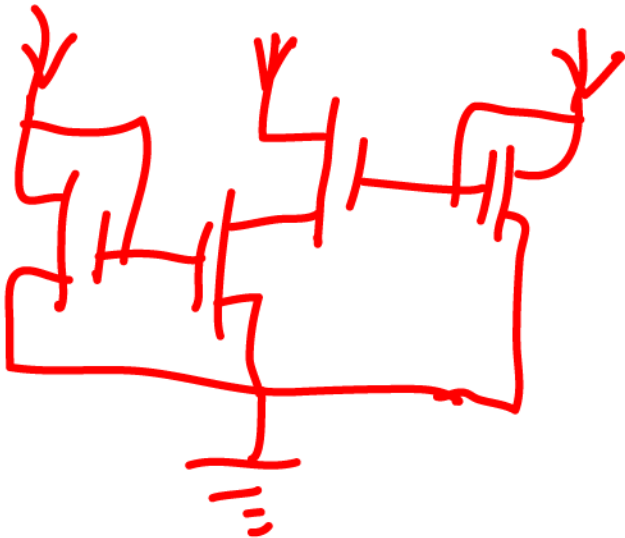
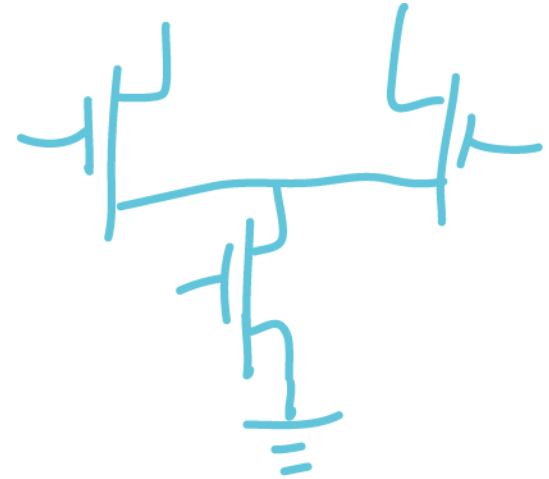
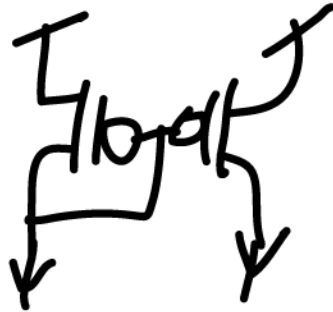


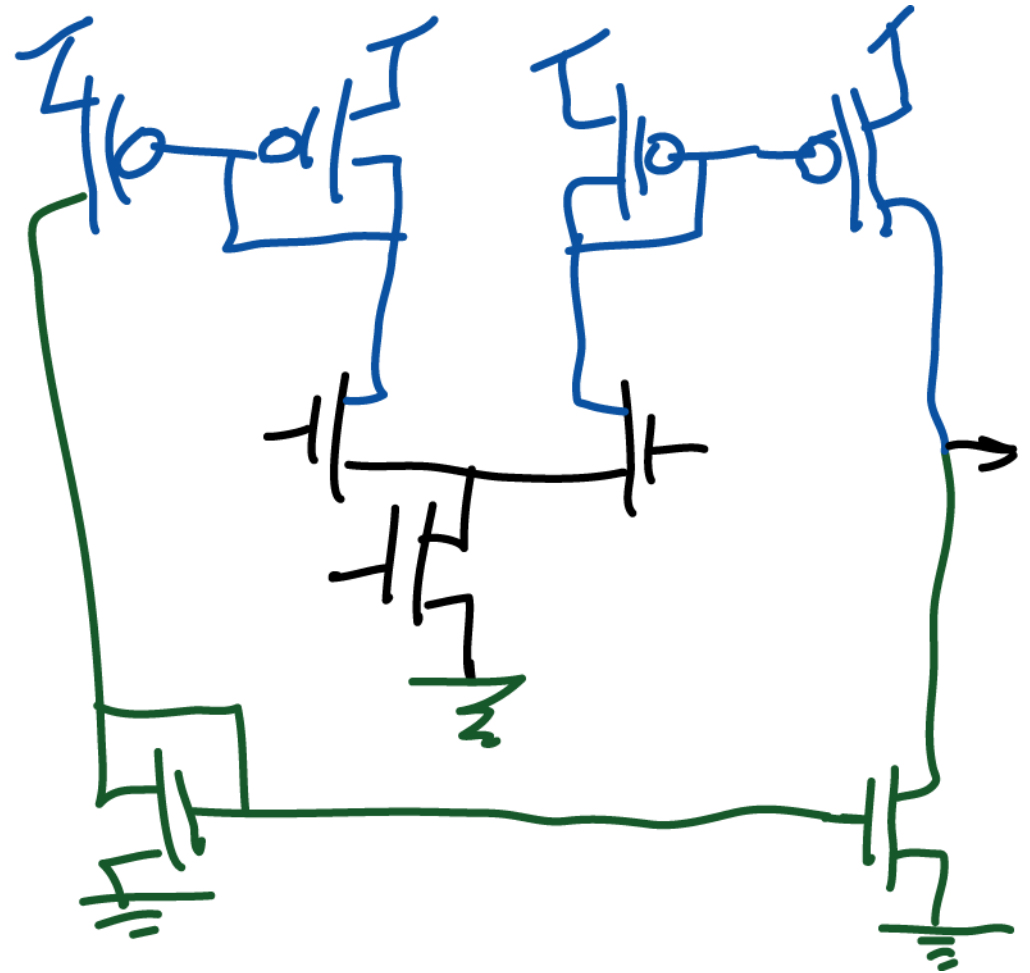
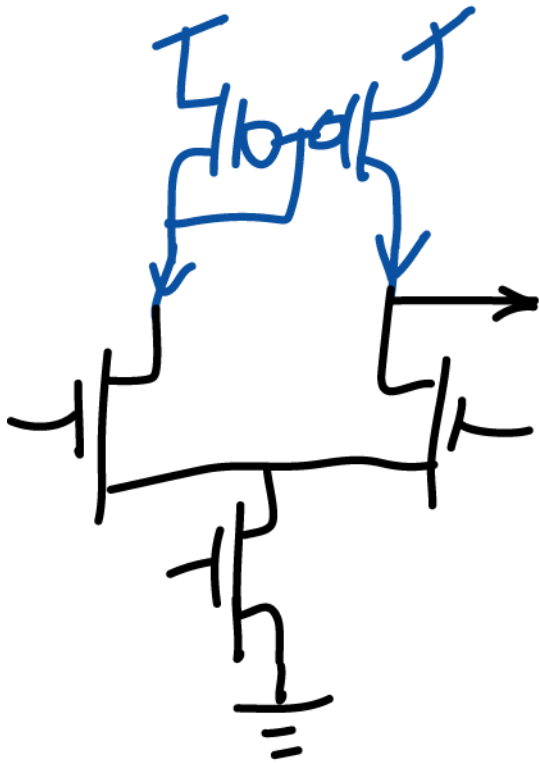
Static Circuits

- Results from Lab3 (Strong inversion+Early)
- Intrinsic gain of a transistor (take 2)
- Diode-connected transistor
- Current mirrors & Scaling (tilted) current mirrors
- Differential pair
- Current correlator
- Bump-Antibump circuit
- Transconductance amplifier (and its g_m and A)
- Wide range transamp

Current mirror, diff-pair, current correlator, bump-antibump circuit



Transconductance amplifier and wide range transconductance amplifier

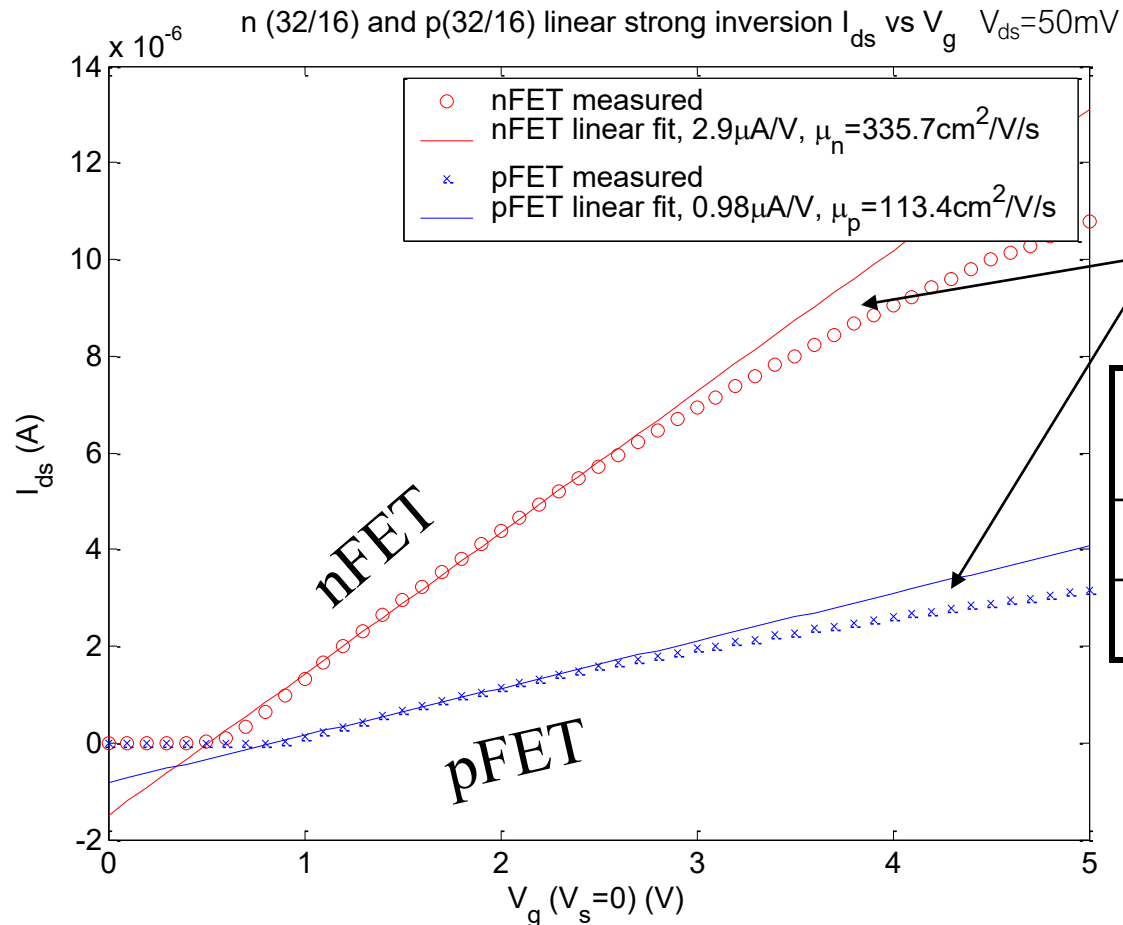


Results from lab 3

Above-threshold transistor
characteristics

How does I_{dlin} scale with $V_g - V_T$?

$$I_{ds} = \mu C_{ox} \frac{W}{L} (V_{gs} - V_T) V_{ds}$$

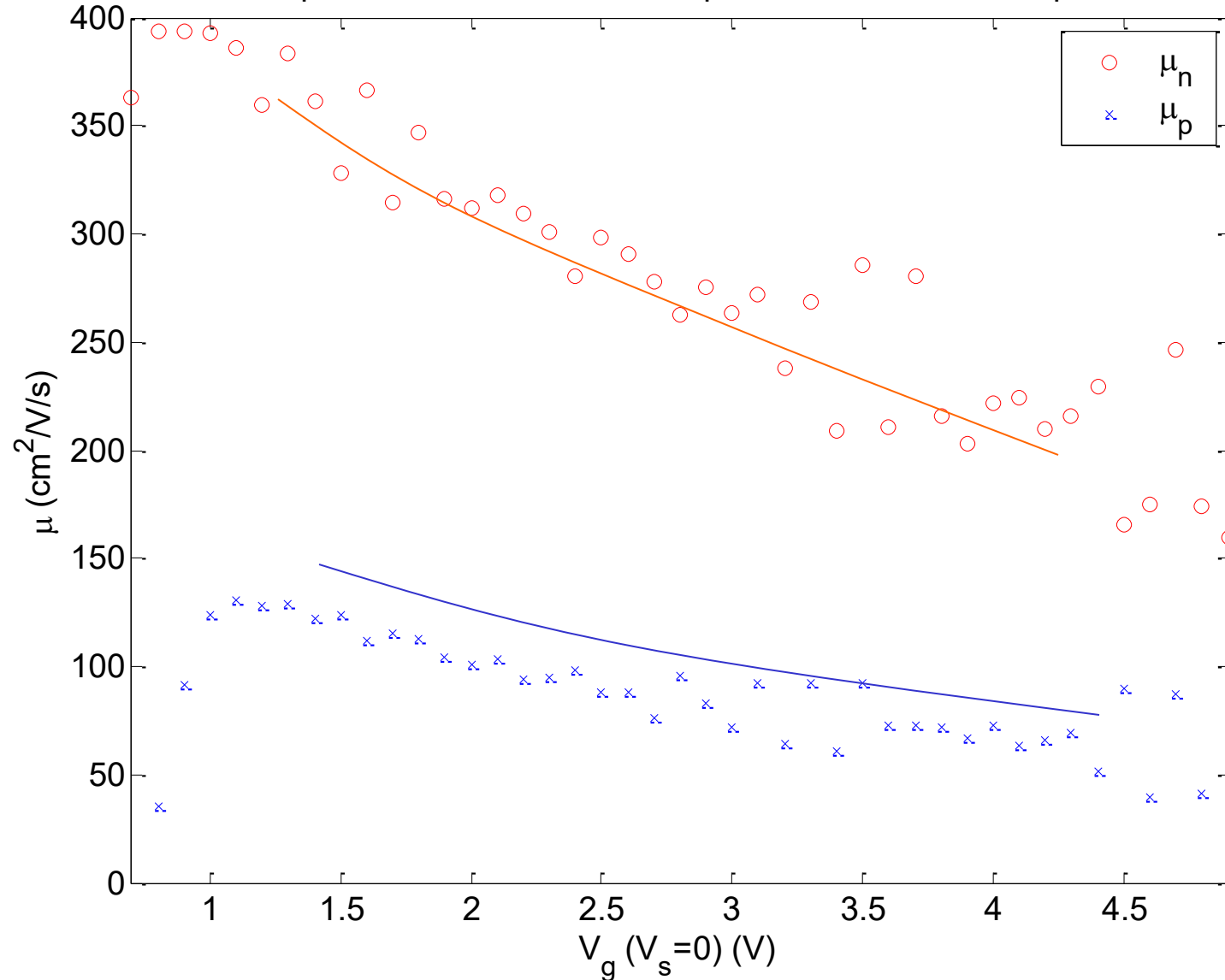


Mobility degradation

μ cm ² /V/s	Meas	Bulk
N	335	1450
P	113	500

What is μ as function of V_g ?

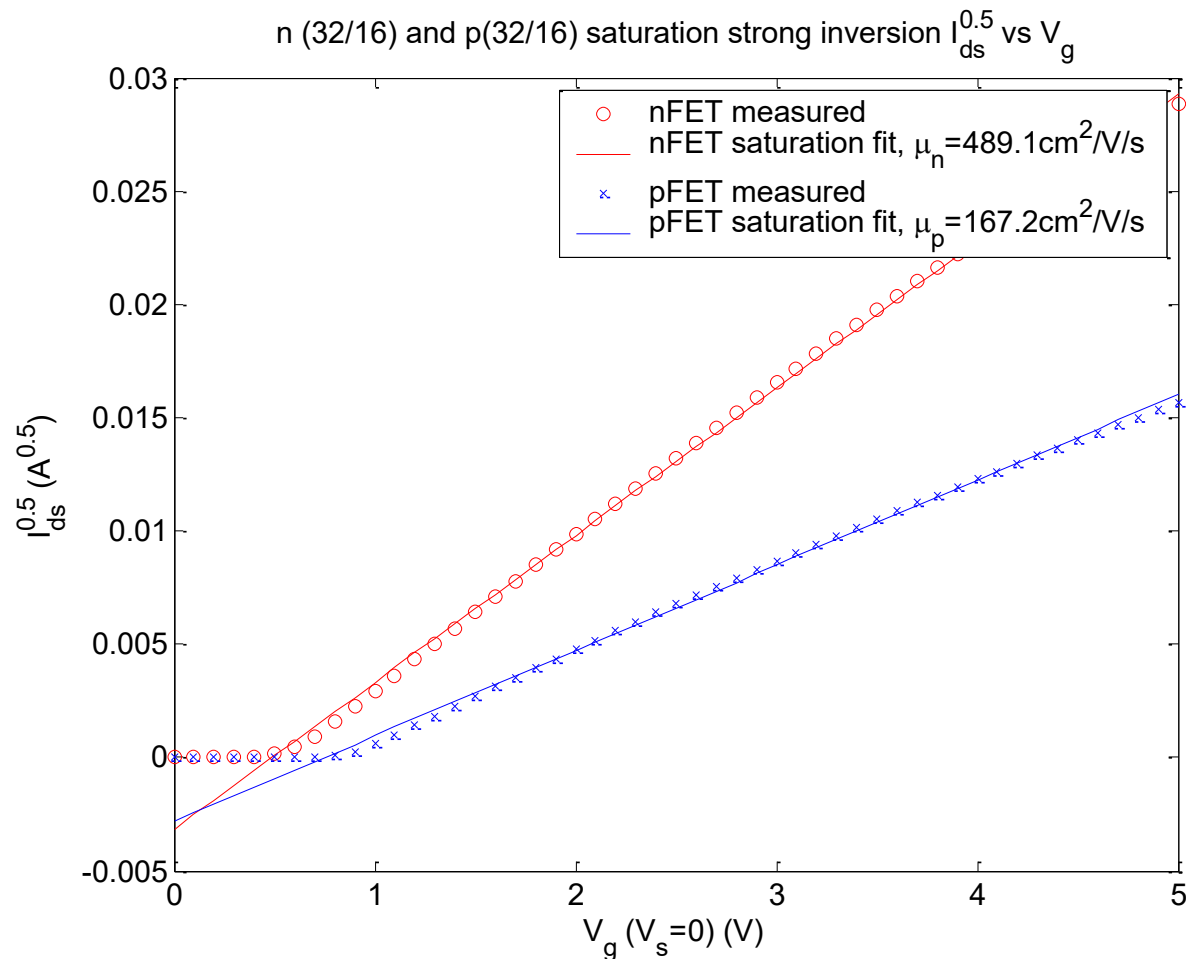
Computed mobilities for ohmic operation of 32/16 n and pfets



How does $\sqrt{I_{dsat}}$ scale with $V_g - V_T$?

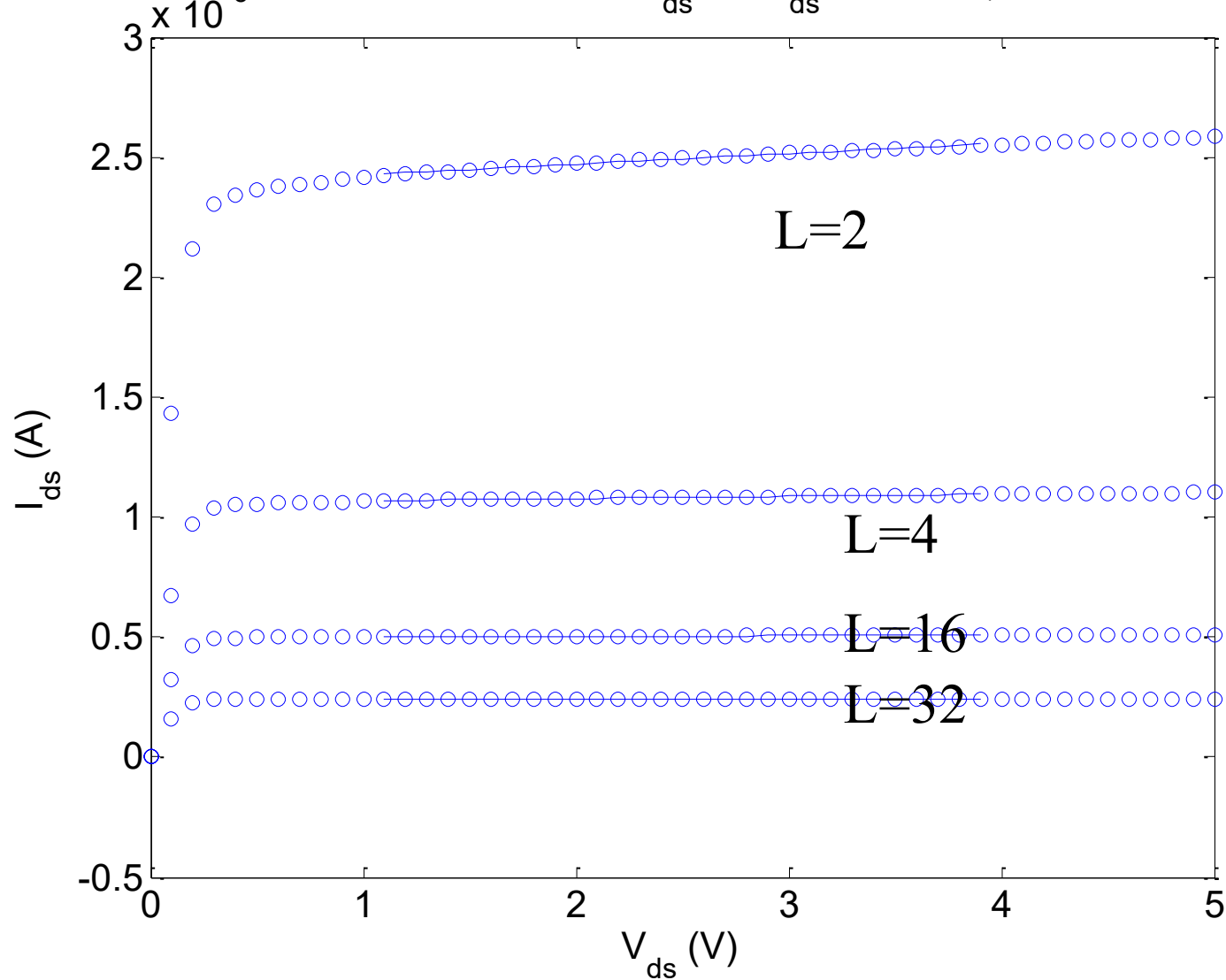
Saturation current

$$\sqrt{I_{ds,sat}} = \sqrt{\frac{\mu C_{ox}}{2} \frac{W}{L}} (V_{gs} - V_T)$$



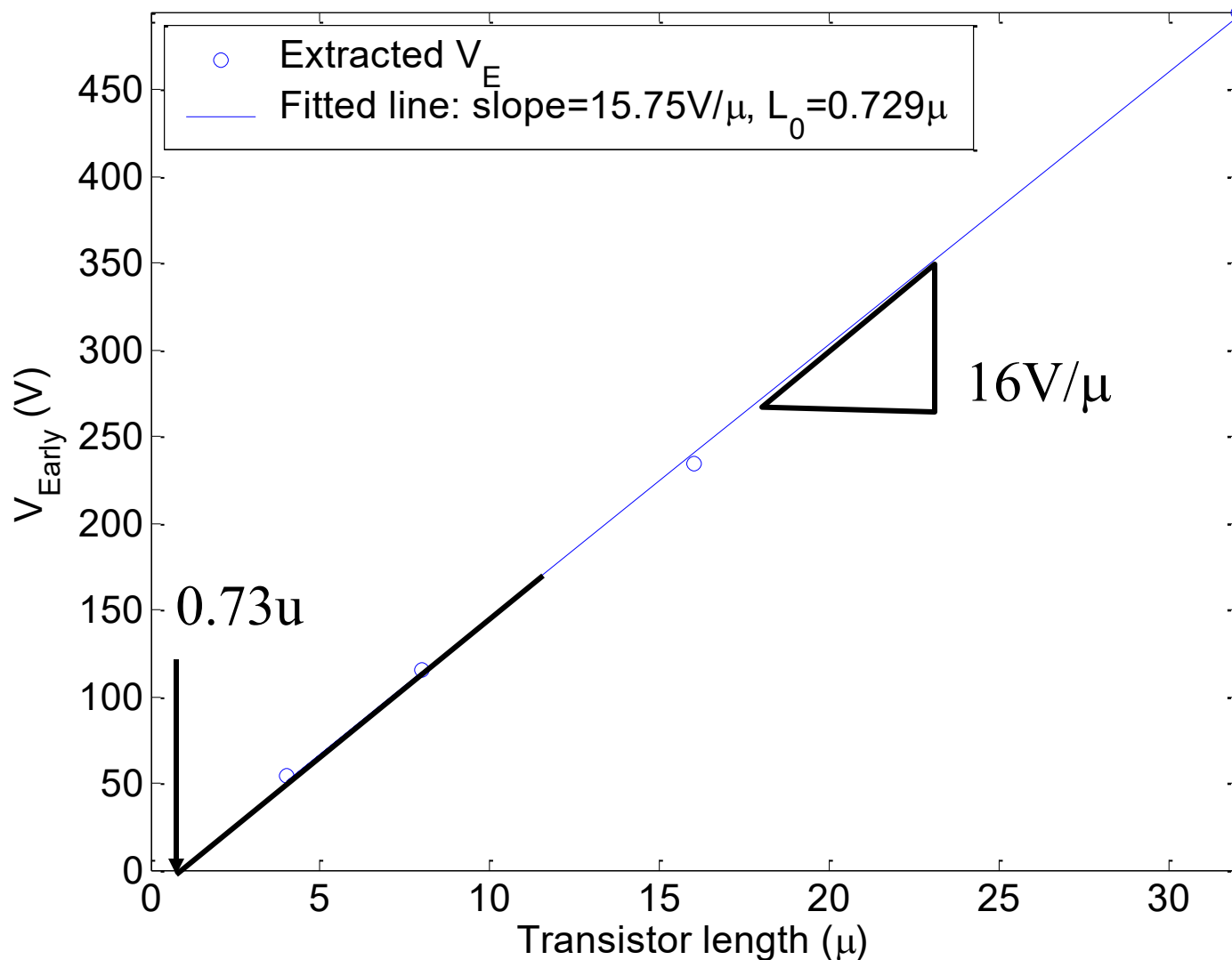
How does I_{ds} scale with L ?

nFET measured above-threshold I_{ds} vs. V_{ds} for $W=32\mu$, $L=2,4,8,16,32\mu$

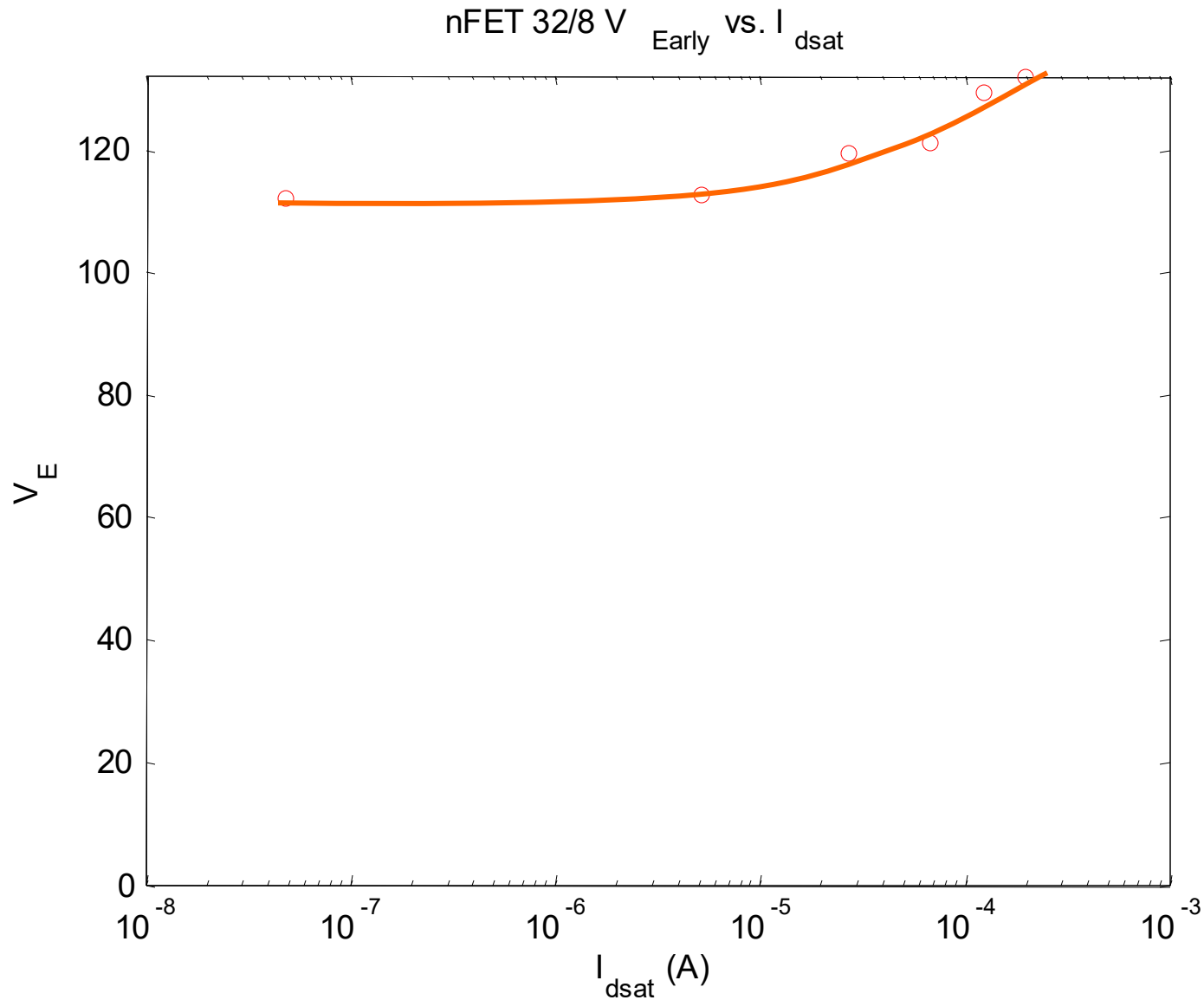


How does V_E change with L ?

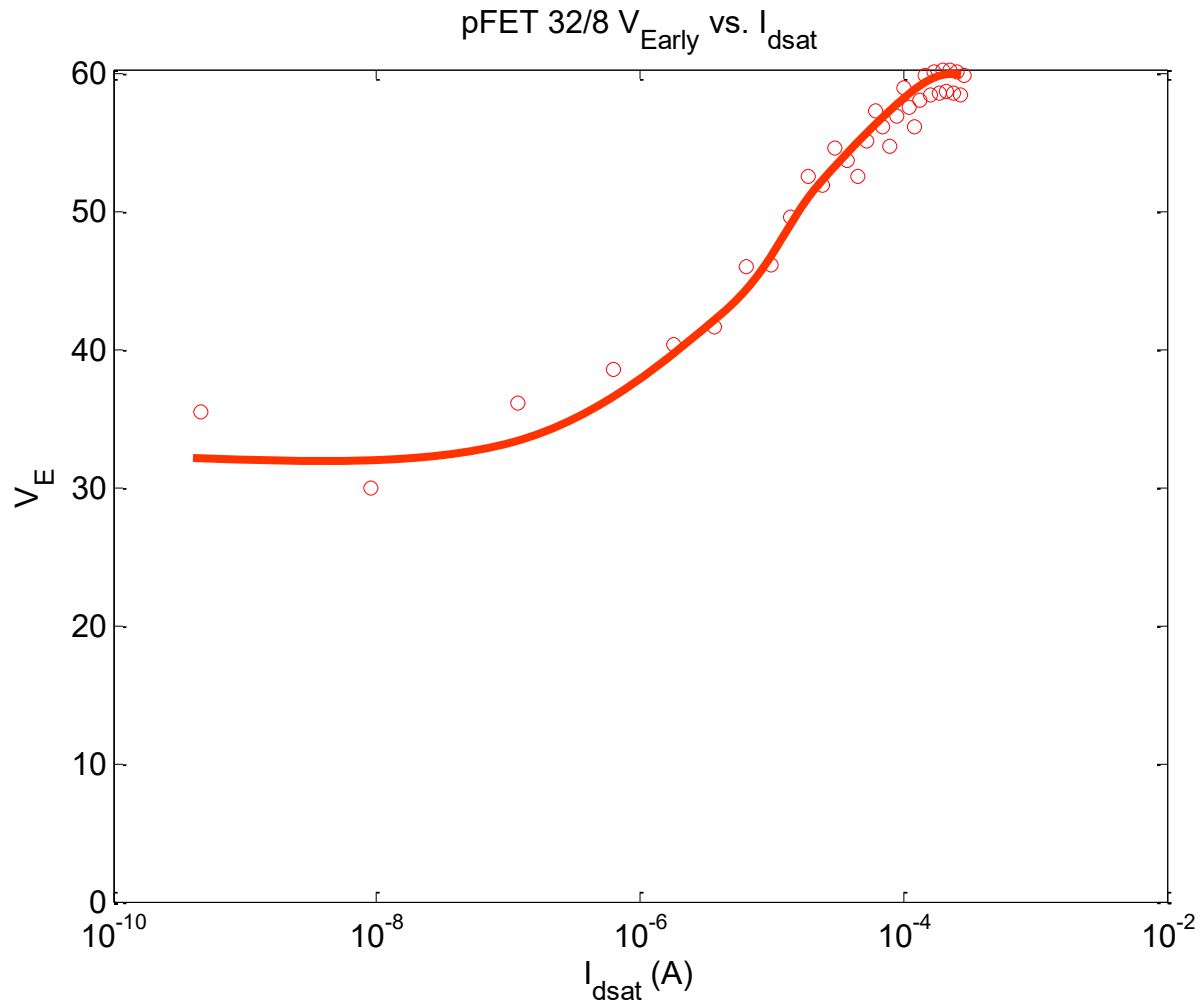
nFET extracted Early voltages for above-threshold operation



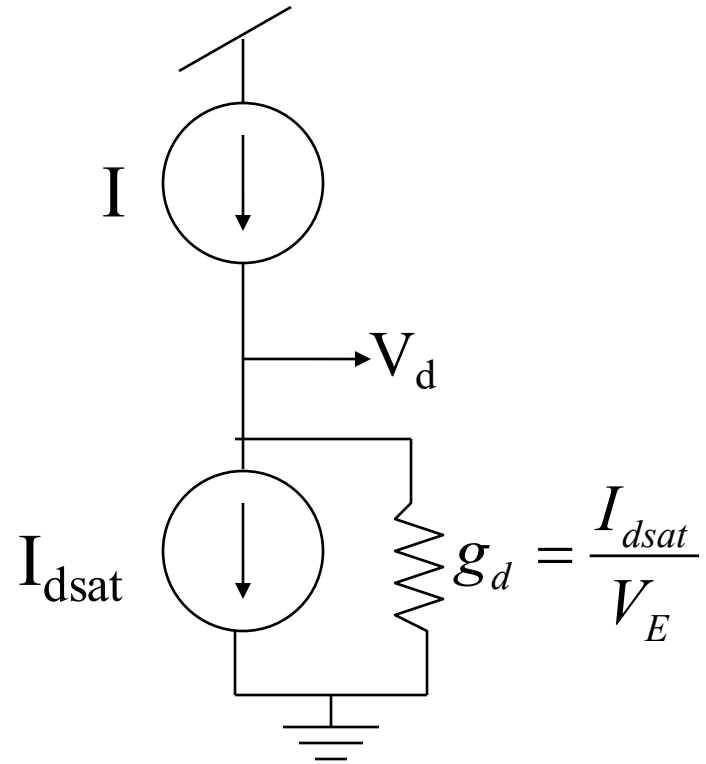
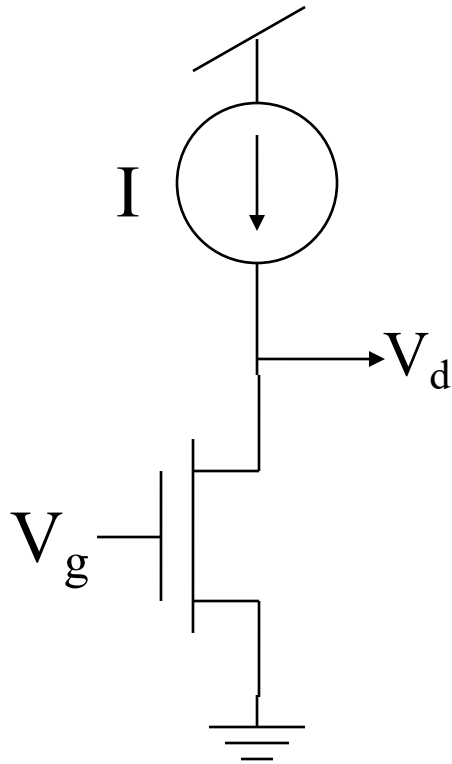
Is V_E constant with I_{dsat} ?



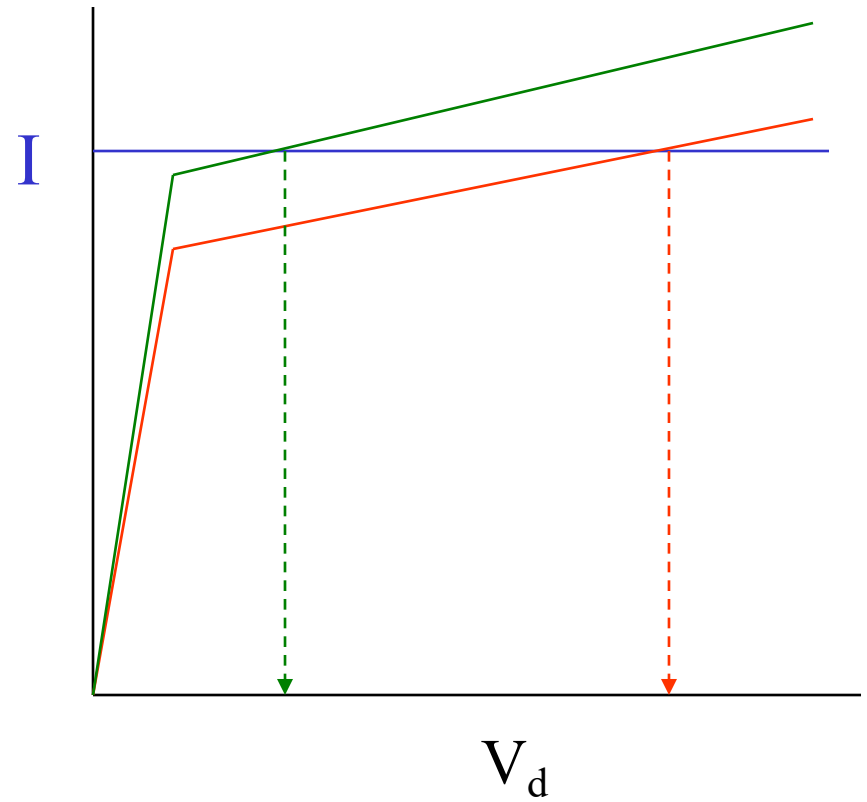
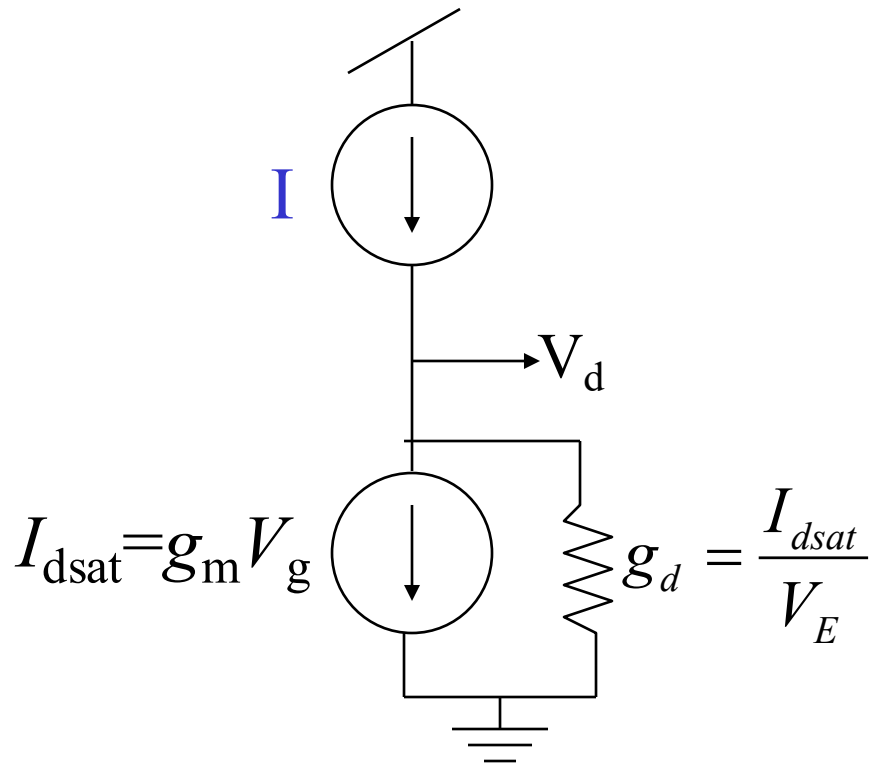
pFET V_E vs. I_{dsat}



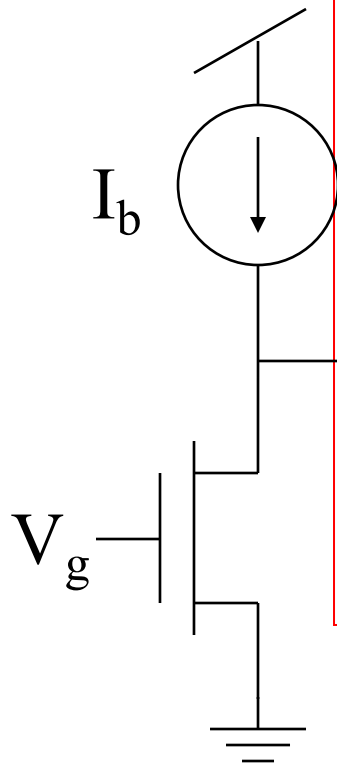
Solution: Intrinsic transistor voltage gain



Intrinsic transistor voltage gain



Intrinsic transistor voltage gain

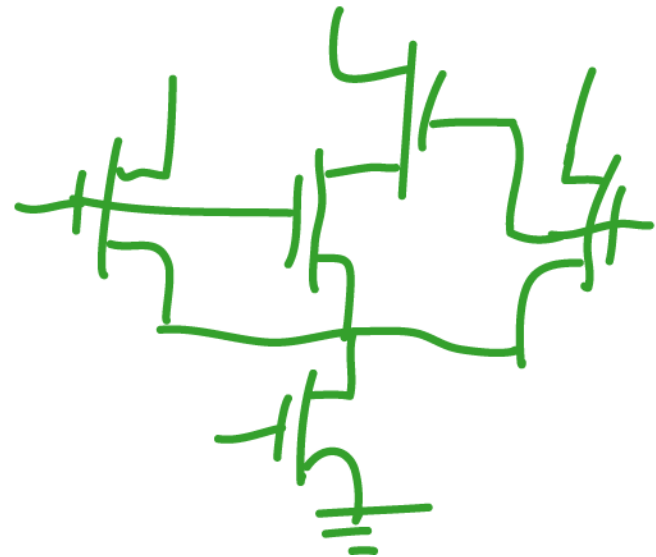
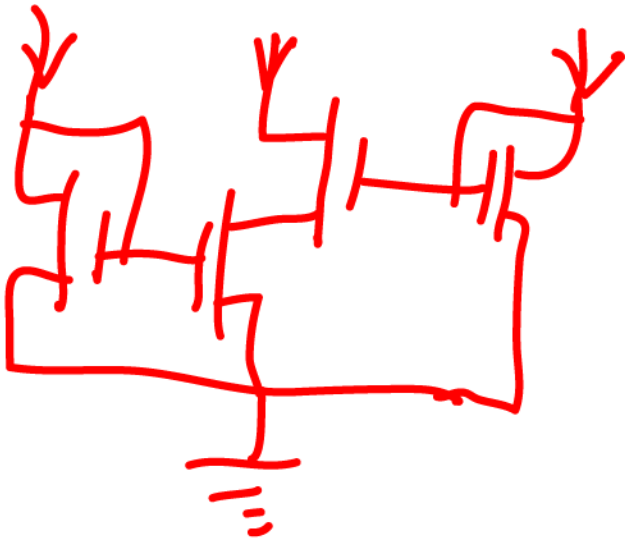
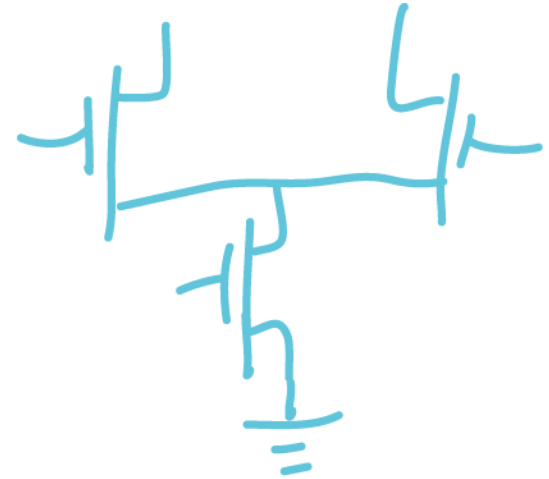
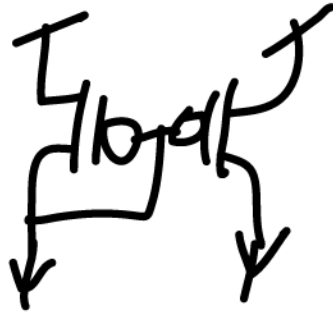


$$\begin{aligned}
 \text{Gain } -A &= \frac{\partial V_d}{\partial V_g} \\
 &= \frac{\partial I}{\partial V_g} \frac{\partial V_d}{\partial I} \\
 &= \frac{g_m}{g_{ds}} = g_m r_o = \frac{\kappa V_E}{U_T} \quad (\text{subthreshold})
 \end{aligned}$$

Typical value: $A = \frac{.75 * 100V}{(1/40)V} = 3000$

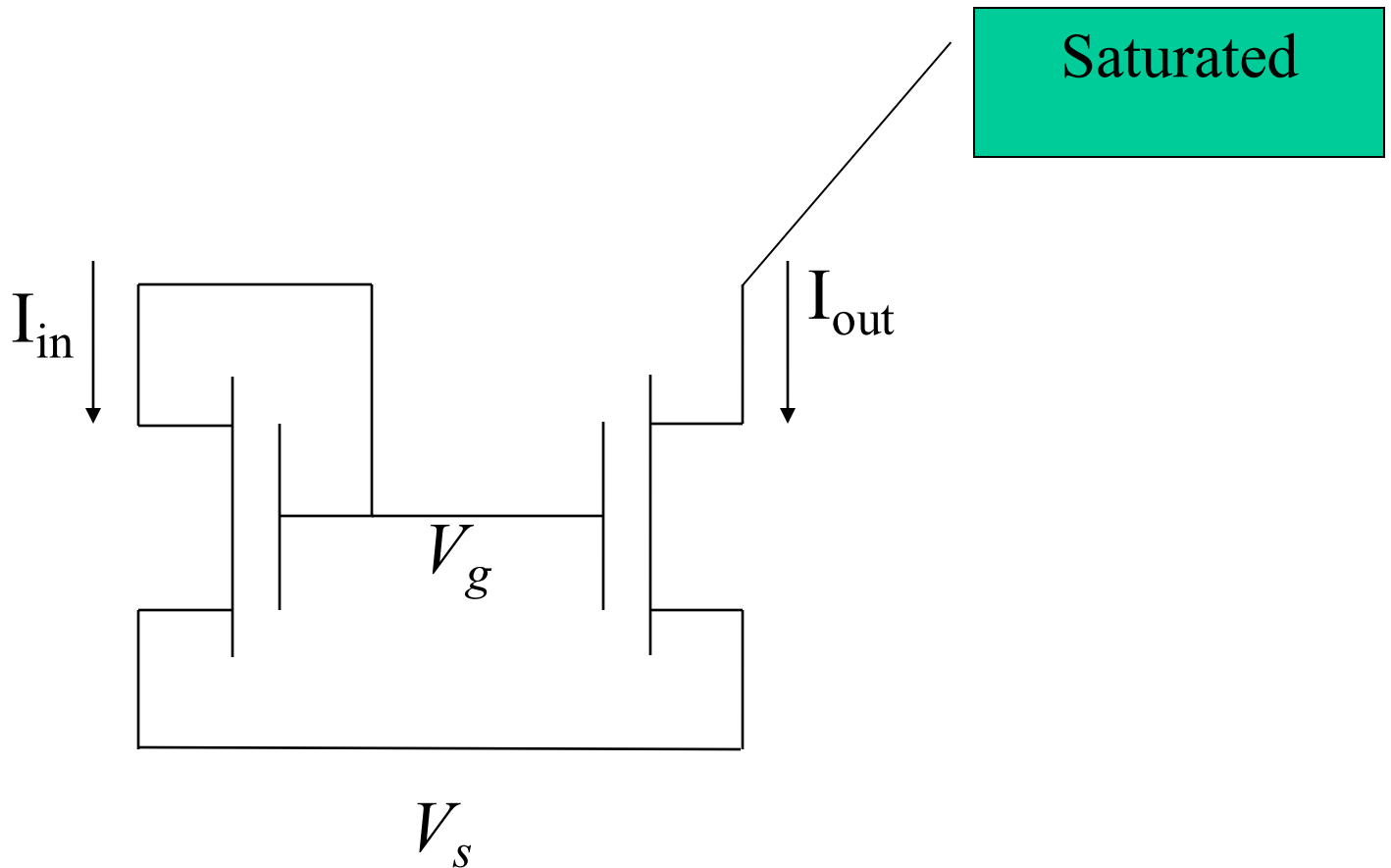
The same equation $A = g_m / g_{out}$ applies to any amplifier, e.g. transamp. g_{out} refers to the total output conductance.

Current mirror, diff-pair, current correlator, bump-antibump circuit

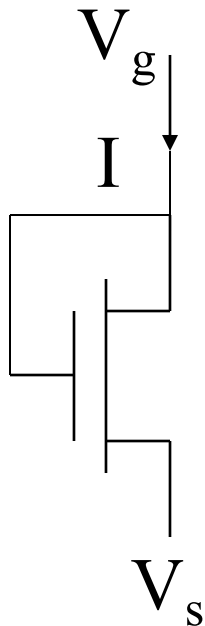


Current Mirror

The output current is a copy of the input current.
It is *mirrored* because it is sunk, not sourced.



Diode-connected transistors



$$\begin{aligned}
 I &= e^{\kappa V_g - V_s} (1 - e^{-(V_d - V_s)}) \\
 &= e^{\kappa V_g - V_s} (1 - e^{-(V_g - V_s)}) \\
 &= e^{\kappa V_g - V_s} - e^{-(1-\kappa)V_g}
 \end{aligned}$$

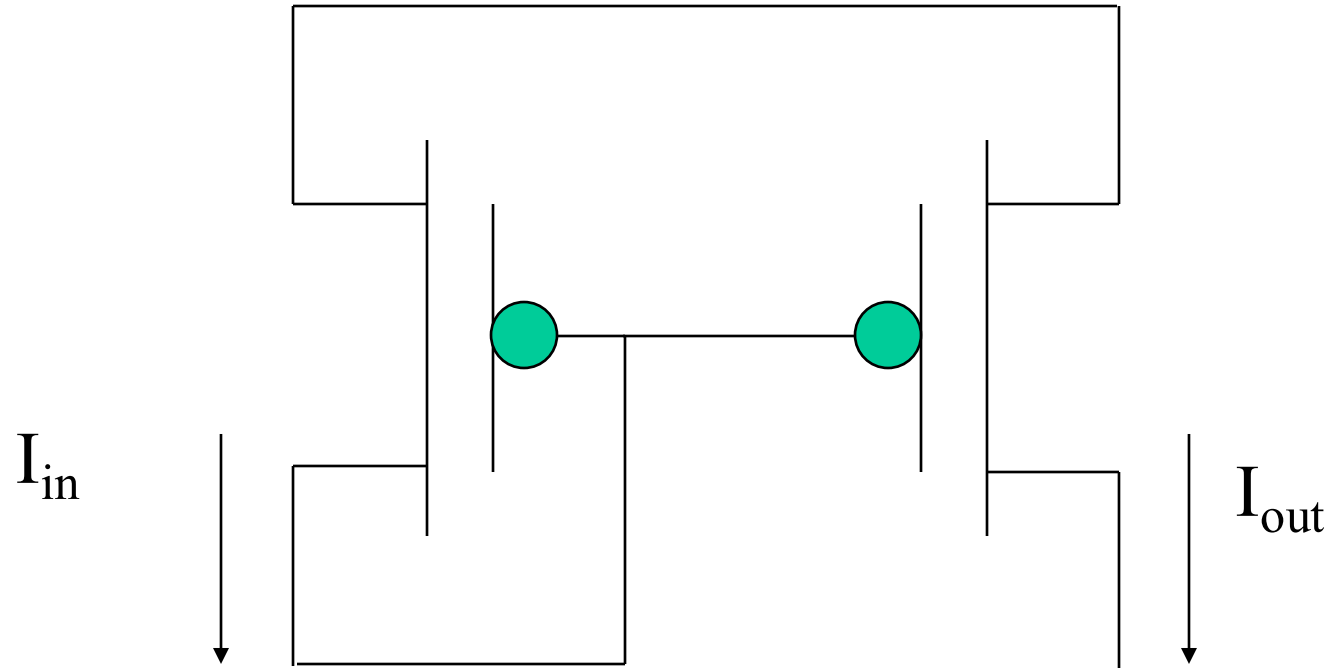
$V_g = V_d$

$$e^{-(1-\kappa)V_g} \rightarrow 0$$

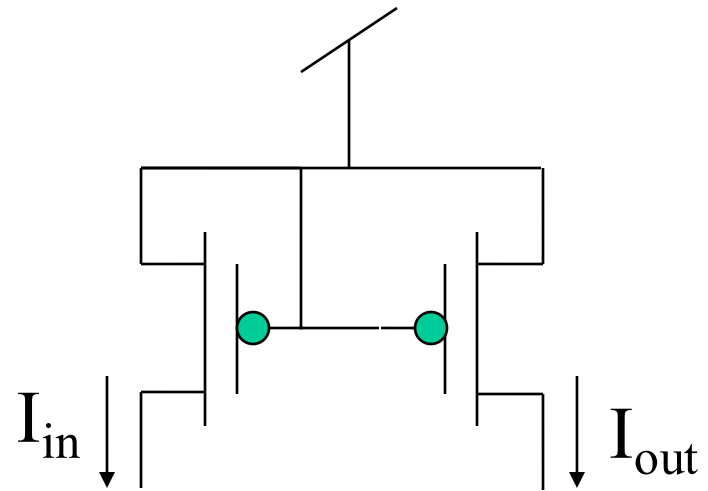
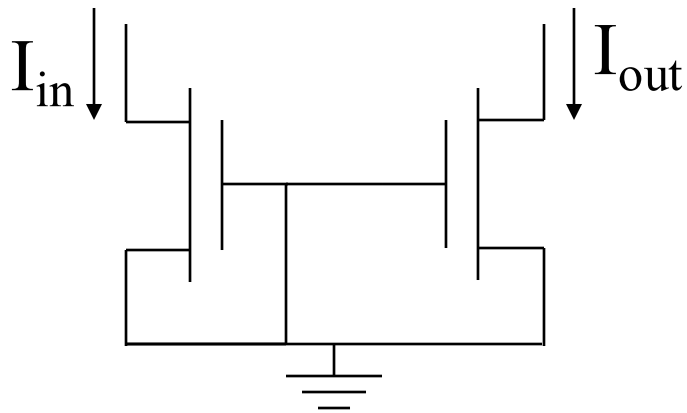
whenever a current more than a few I_0 flows

$$I = e^{\kappa V_g - V_s}$$

pFET mirror



How about these configurations?

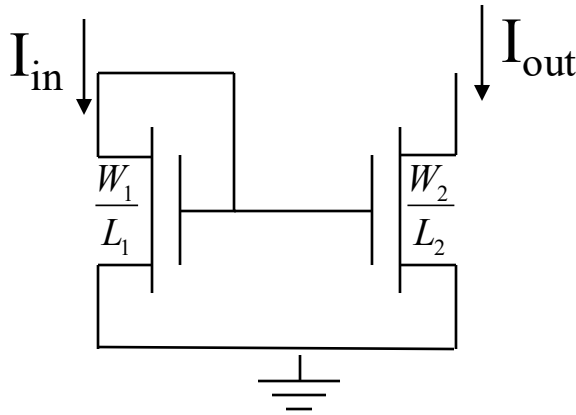


$$V_{gs}=0, \text{ so } I_{out}=I_0$$

Current mirror with gain (tilted mirror) (I)

How do you make an output current that is M times the input current?

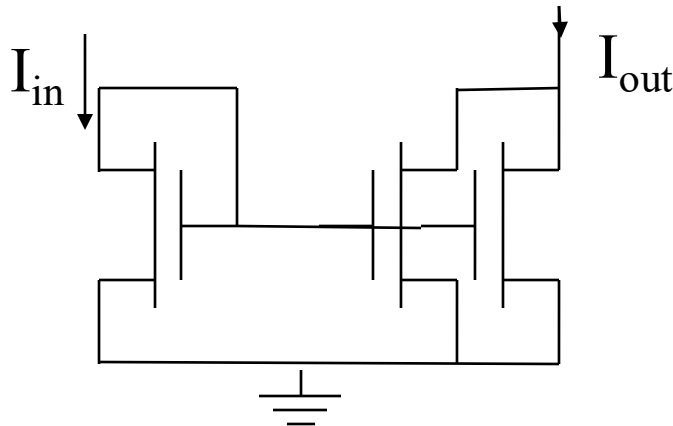
METHOD 1: USE TRANSISTOR GEOMETRY



$$\text{Gain } M = \frac{W_2/L_2}{W_1/L_1}$$

Not very accurate when $M \neq 1$

METHOD 2: USE MULTIPLE TRANSISTORS



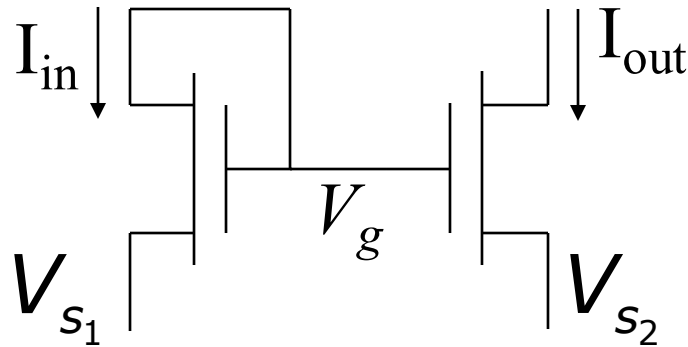
$$\text{e.g. Gain } M = 2$$

More accurate

$$\text{Precision} \propto \frac{1}{\sqrt{WL}}$$

Current mirror with gain (tilted mirror) (II)

METHOD 3: USE A DIFFERENT V_s



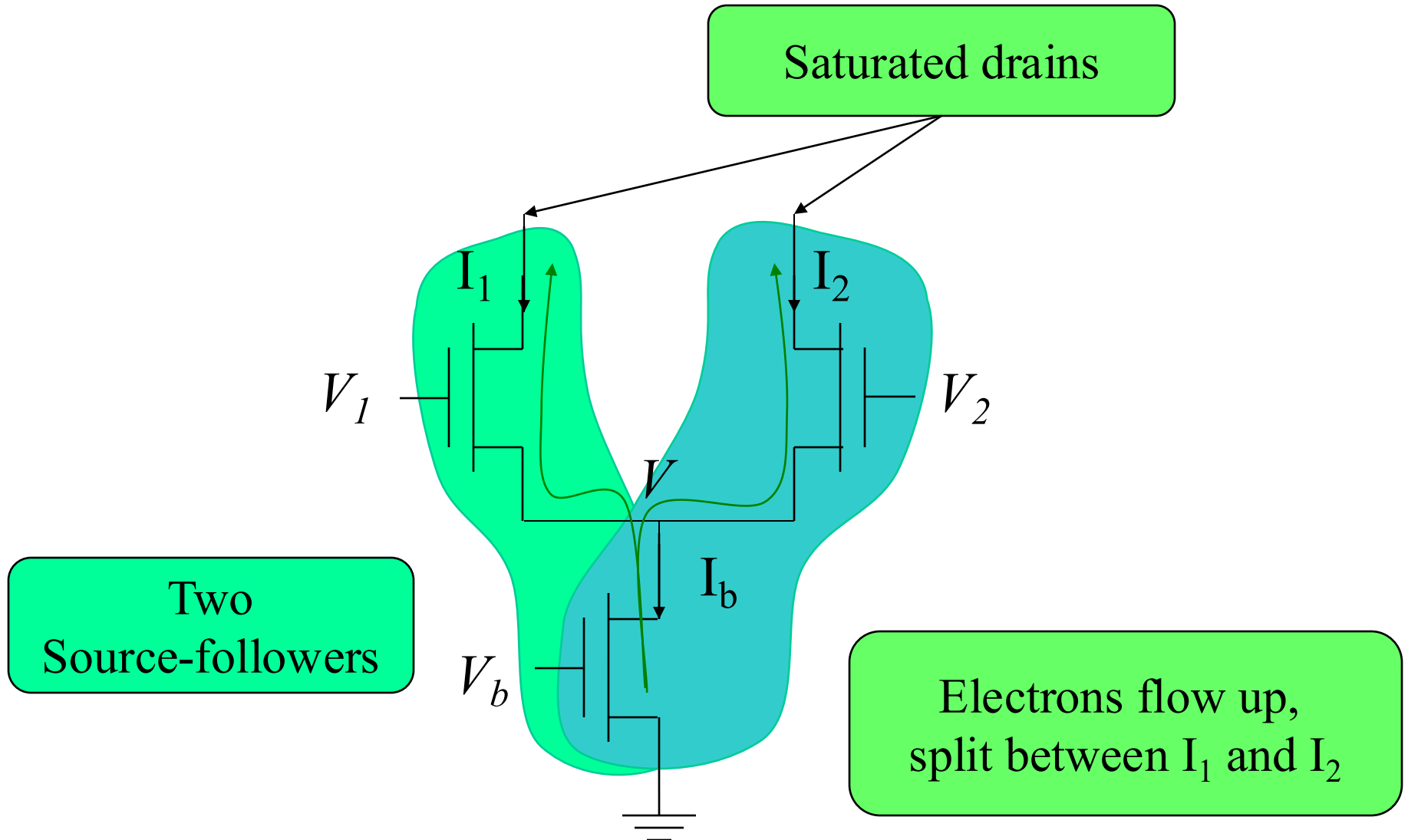
$$I_{in} = e^{\kappa V_g - V_{s1}}$$

$$I_{out} = e^{\kappa V_g - V_{s2}}$$

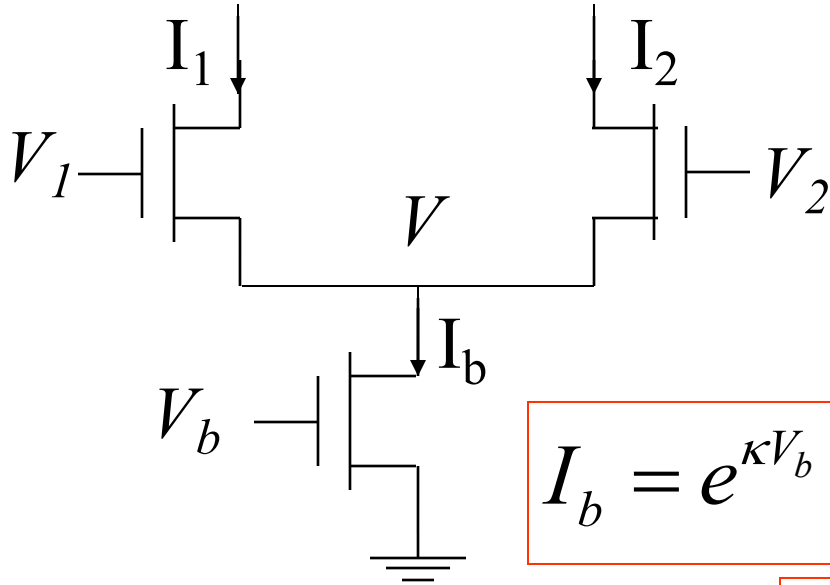
$$[V] = U_T$$

$$\text{Gain } M = \frac{I_2}{I_1} = e^{V_{s1} - V_{s2}}$$

Differential Pair (I)



Differential Pair (II)



$$I_1 = I_0 e^{\kappa V_1 - V}$$

$$I_2 = I_0 e^{\kappa V_2 - V}$$

$$I_b = e^{\kappa V_b} = I_1 + I_2 = I_0 e^{-V} (e^{\kappa V_1} + e^{\kappa V_2})$$

$$\Rightarrow e^V = \frac{e^{\kappa V_1} + e^{\kappa V_2}}{e^{\kappa V_b}}$$

$$V = \ln (e^{\kappa V_1} + e^{\kappa V_2}) - \kappa V_b$$

$$\approx \kappa (V_1 - V_b) \text{ for } V_1 - V_2 \geq 100\text{mV}$$

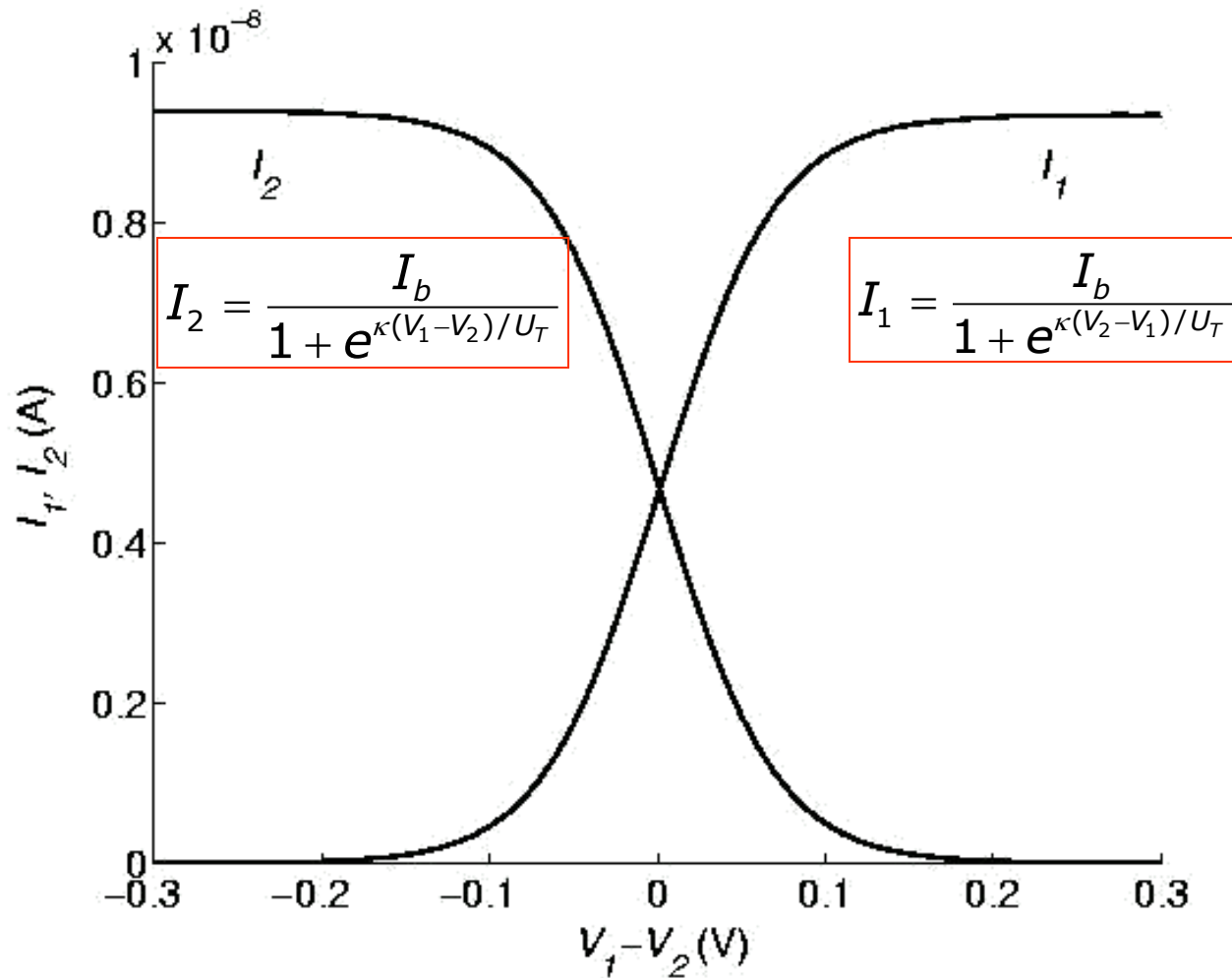
Differential Pair (III)

$$I_1 = \frac{I_b e^{\kappa V_1}}{e^{\kappa V_1} + e^{\kappa V_2}}$$
$$= \frac{I_b}{1 + e^{\kappa(V_2 - V_1)}}$$

$$I_2 = \frac{I_b}{1 + e^{\kappa(V_1 - V_2)}}$$

FERMI
FUNCTIONS

Differential Pair (IV)



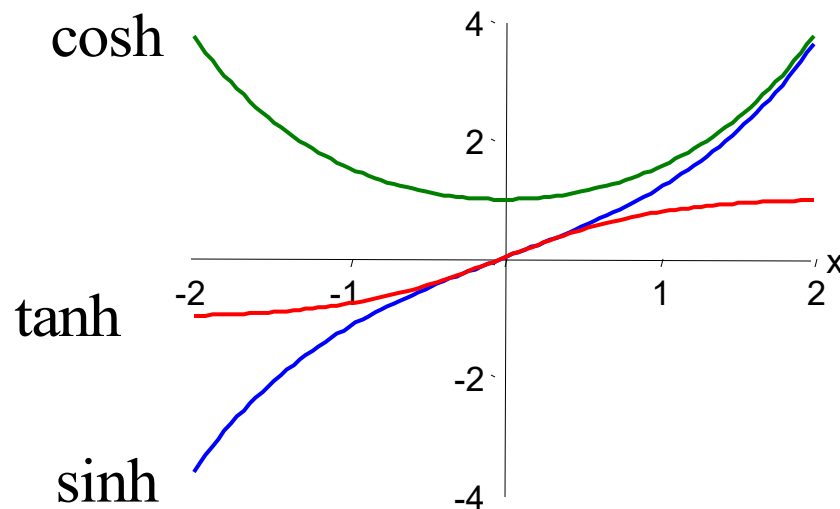
$$I_1 - I_2 = I_b \frac{e^{\kappa V_1} - e^{\kappa V_2}}{e^{\kappa V_1} + e^{\kappa V_2}} = I_b \tanh \left(\kappa \frac{V_1 - V_2}{2} \right)$$

Digression: Hyperbolic functions

$$\sinh(x) = \frac{e^x - e^{-x}}{2} \approx x \text{ for small } x$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2} \approx 1 + \frac{x^2}{2} \text{ for small } x$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \approx x \text{ for small } x$$



tanh: compressive
sinh: expansive

Differential pair in strong inversion (V)

• In **strong inversion**, solve for I_1 & I_2 this way:

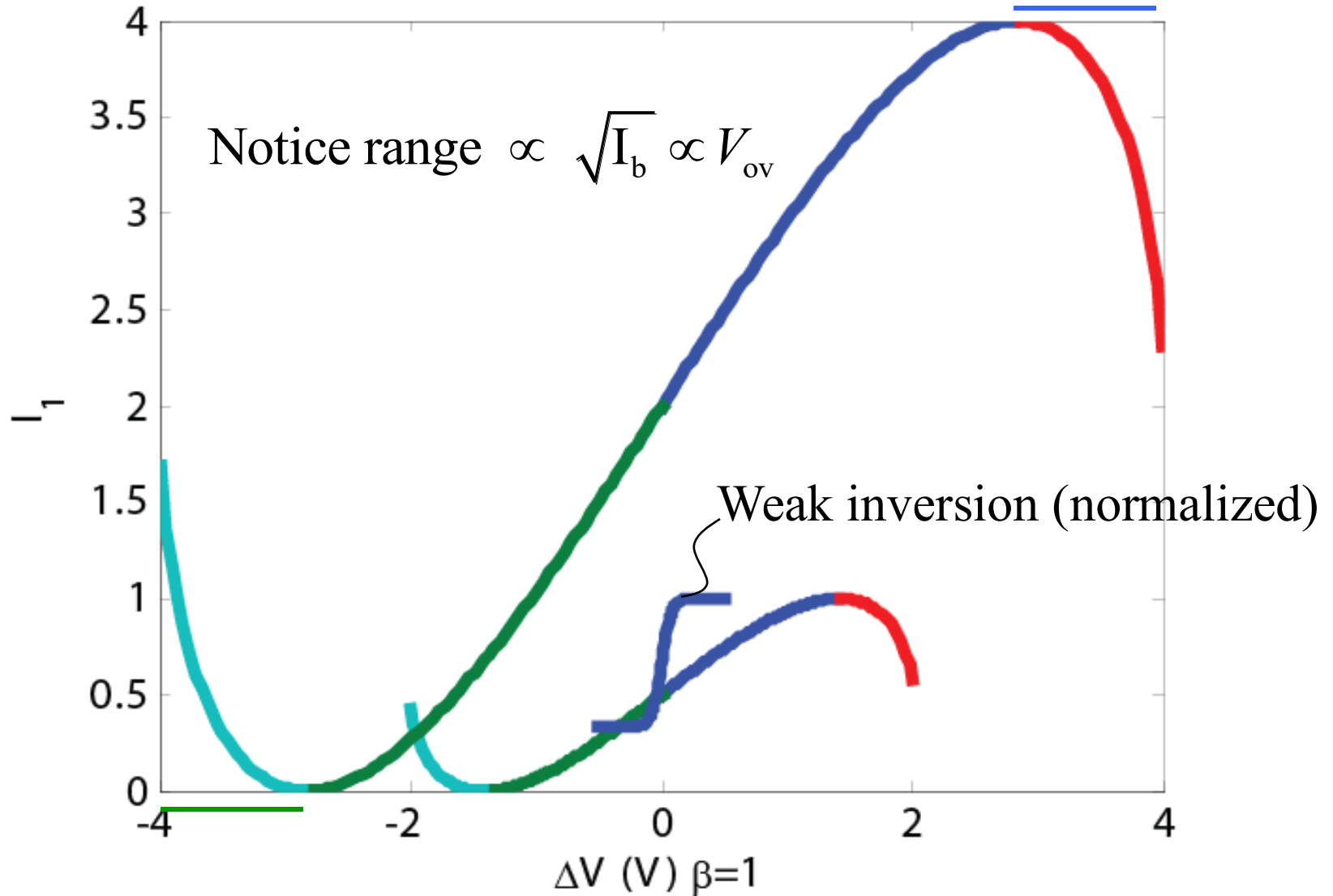
$$\Delta V \equiv V_1 - V_2 = \sqrt{\frac{2I_1}{\beta}} - \sqrt{\frac{2I_2}{\beta}}$$

You eventually obtain

$$\text{for } \frac{\beta \Delta V^2}{2} < I_b \quad I_{1,2} = \frac{I_b}{2} \left(1 \begin{matrix} + \\ - \end{matrix} \sqrt{\frac{\beta \Delta V^2}{I_b} - \left(\frac{\beta \Delta V^2 / 2}{I_b} \right)^2} \right)$$

$$I_1 - I_2 = I_b \sqrt{\frac{\beta \Delta V^2}{I_b} - \left(\frac{\beta \Delta V^2 / 2}{I_b} \right)^2}$$

Differential pair in weak and strong inversion



Differential pair transconductance (VII)

Transconductance of differential tail current

$$g_m = \frac{d(I_1 - I_2)}{d(V_1 - V_2)}$$

Weak inversion

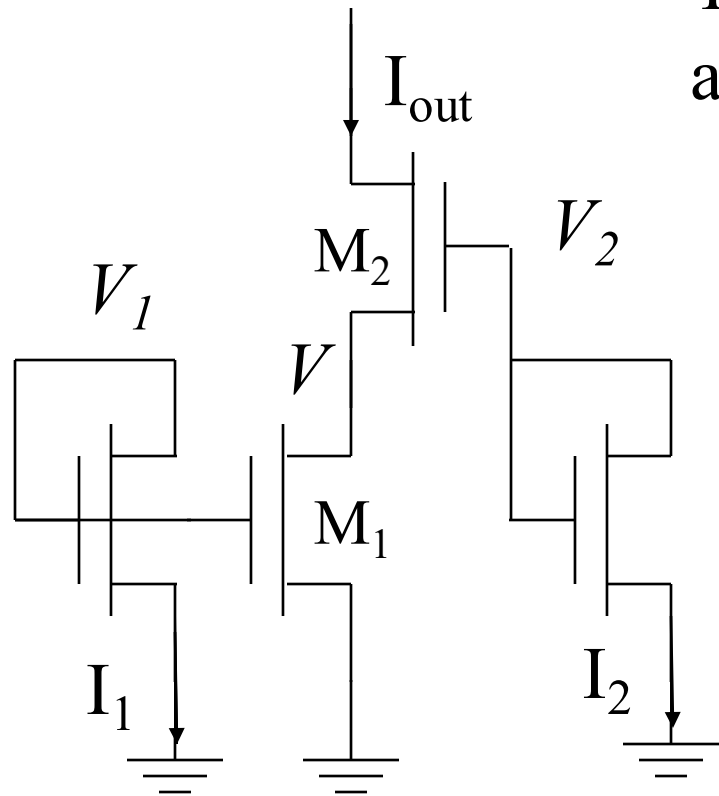
$$g_m = \frac{\kappa I_b}{2U_T}$$

Strong inversion

$$\begin{aligned} g_m &= \sqrt{\beta I_b} \\ &= \sqrt{2} \frac{I_b}{V_b - V_T} \end{aligned}$$

Current Correlator

Transistor M_1 is in ohmic region and transistor M_2 is in saturation.



$$I_{out} = I_0 e^{\kappa V_1} (1 - e^V) \quad (M_1)$$

$$I_{out} = I_0 e^{\kappa V_2 - V} \quad (M_2)$$

$$e^V = \frac{I_0 e^{\kappa V_2}}{I_{out}}$$

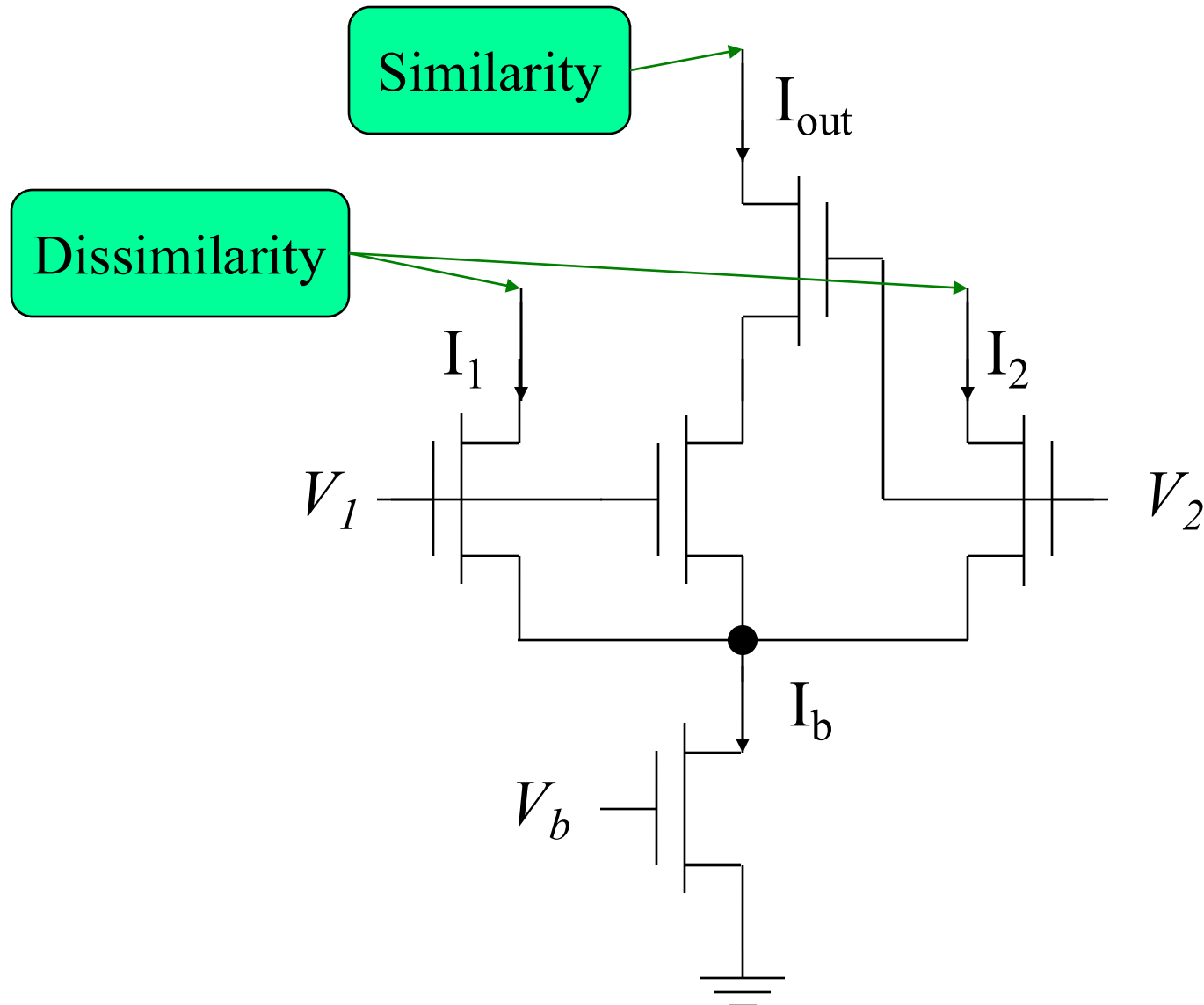
$$I_{out} = \frac{I_1 I_2}{I_1 + I_2}$$

Self-normalized product

$$I_1 = I_0 e^{\kappa V_1}$$

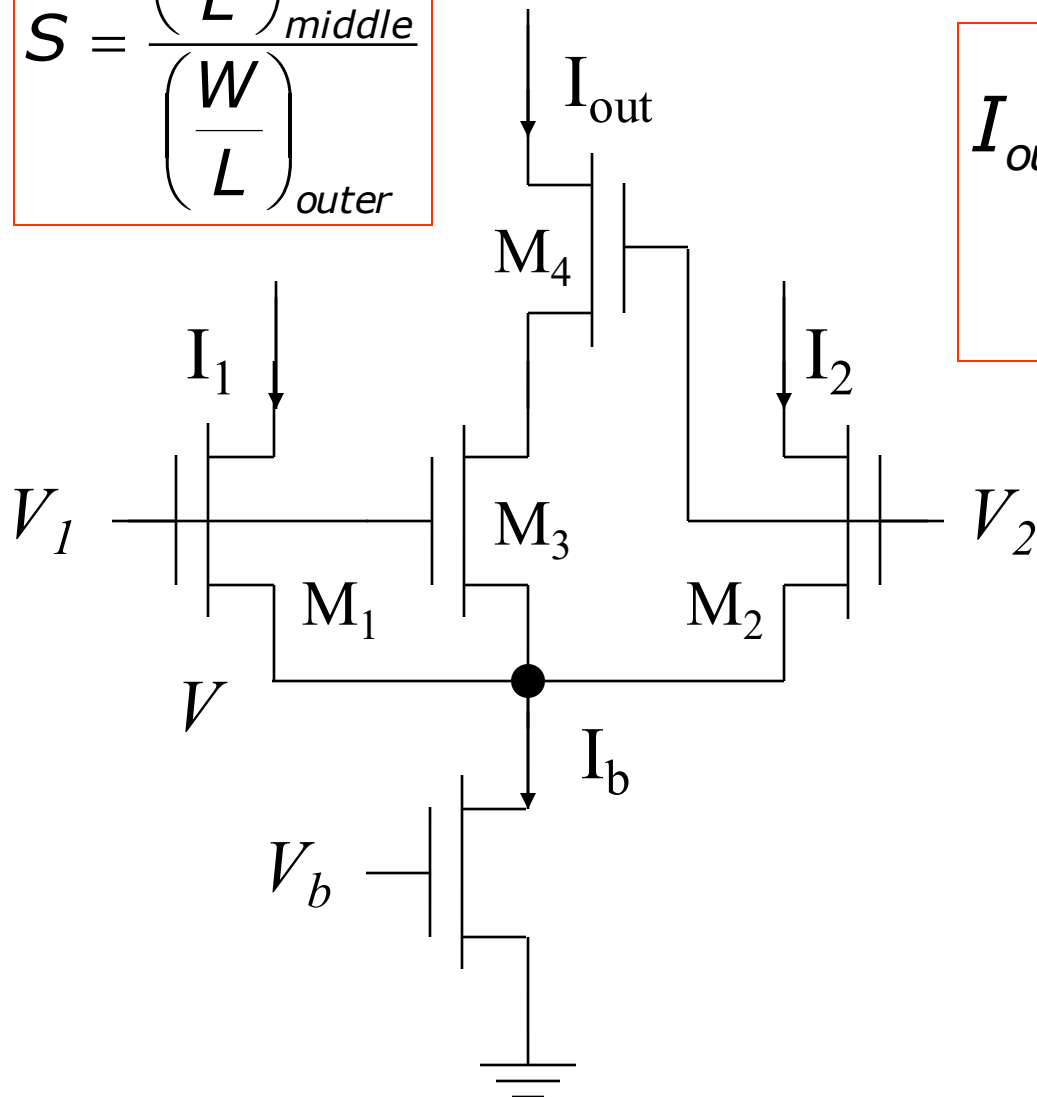
$$I_2 = I_0 e^{\kappa V_2}$$

Bump-Antibump circuit



$$S = \frac{\left(\frac{W}{L}\right)_{middle}}{\left(\frac{W}{L}\right)_{outer}}$$

Bump Circuit

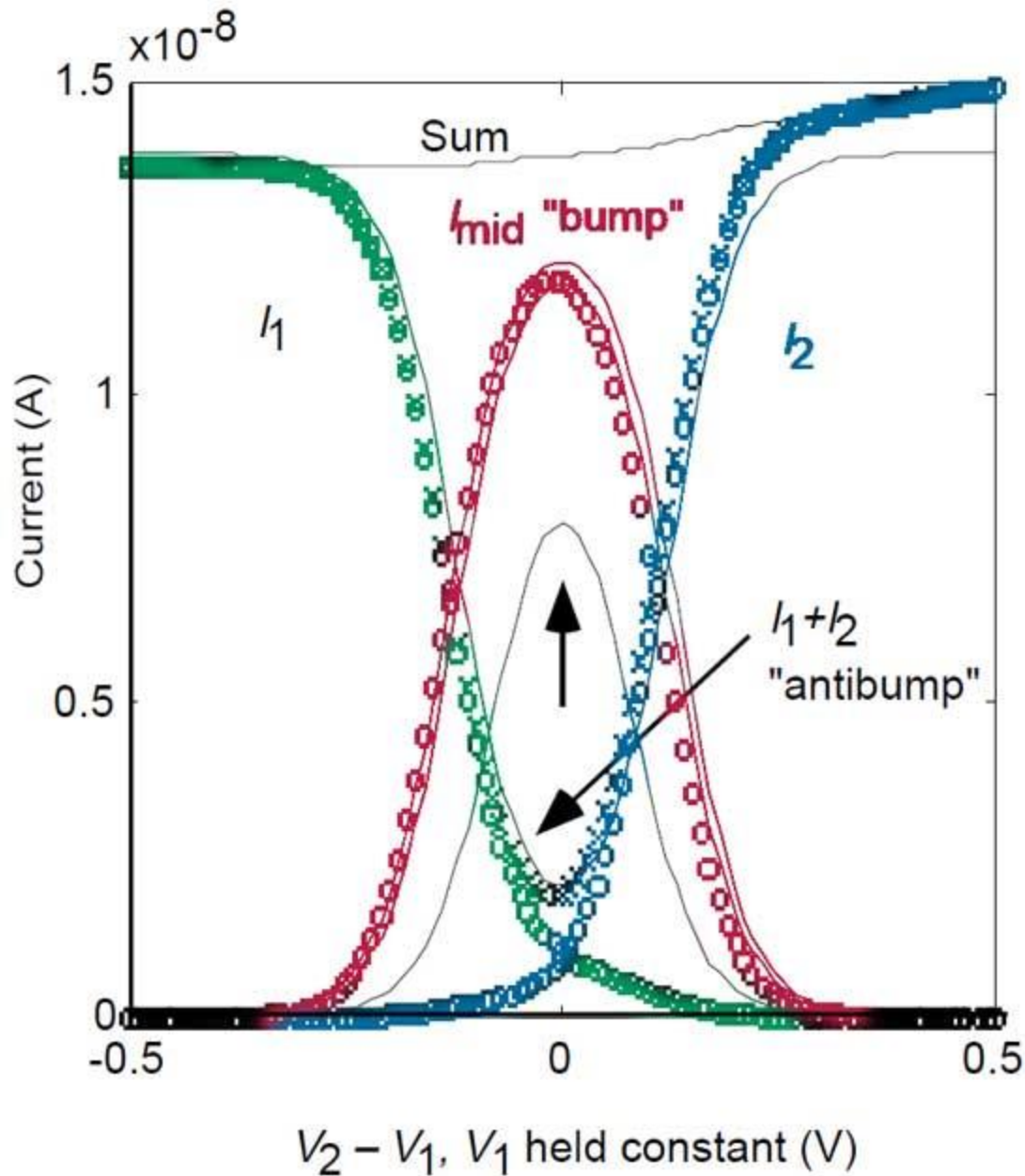
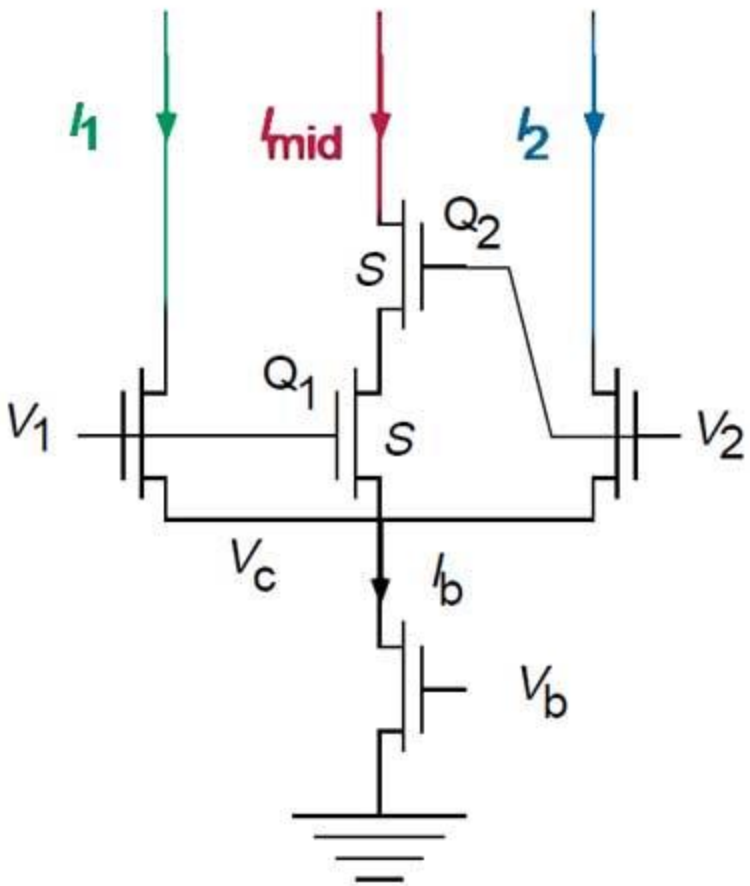


$$I_{out} = \frac{I_b}{1 + \frac{4}{S} \cosh^2\left(\frac{\kappa \Delta V}{2U_T}\right)}$$

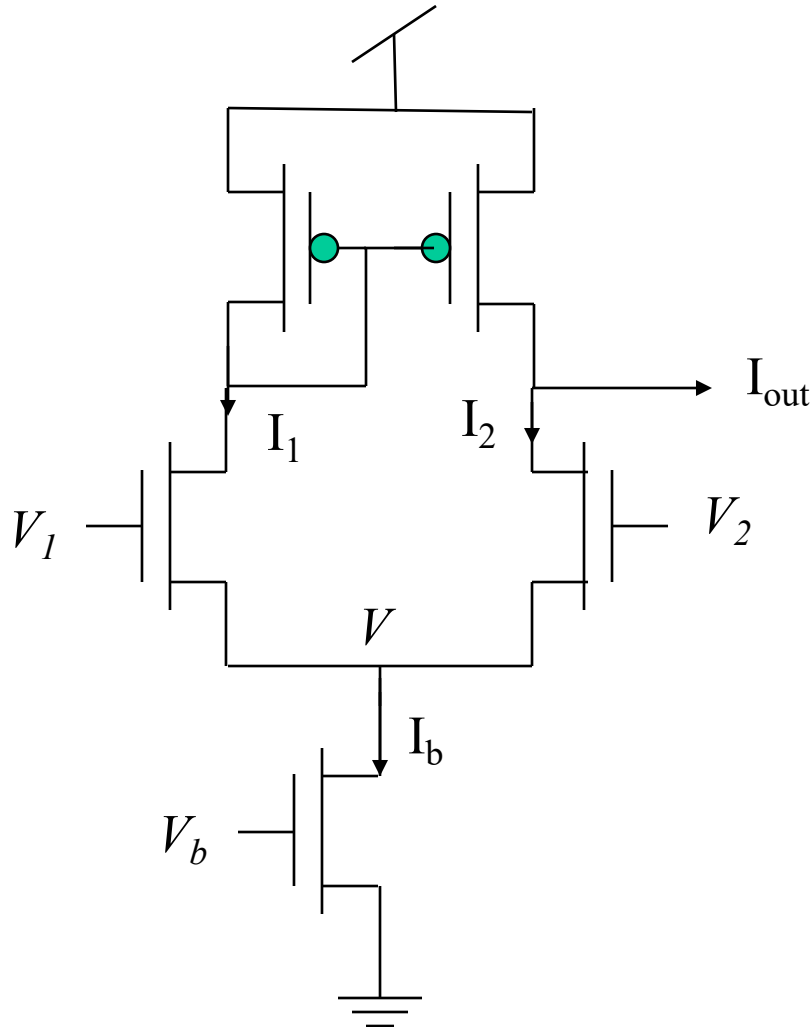
$$\begin{aligned} I_1 + I_2 &= I_b - I_{out} \\ &= \frac{I_b}{\frac{S}{4} \cosh^{-2}\left(\frac{\kappa \Delta V}{2U_T}\right) + 1} \end{aligned}$$

$$(I_1 + I_2)_{\min} = \frac{I_b}{\frac{S}{4} + 1}$$

Bump Circuit



Transconductance amplifier

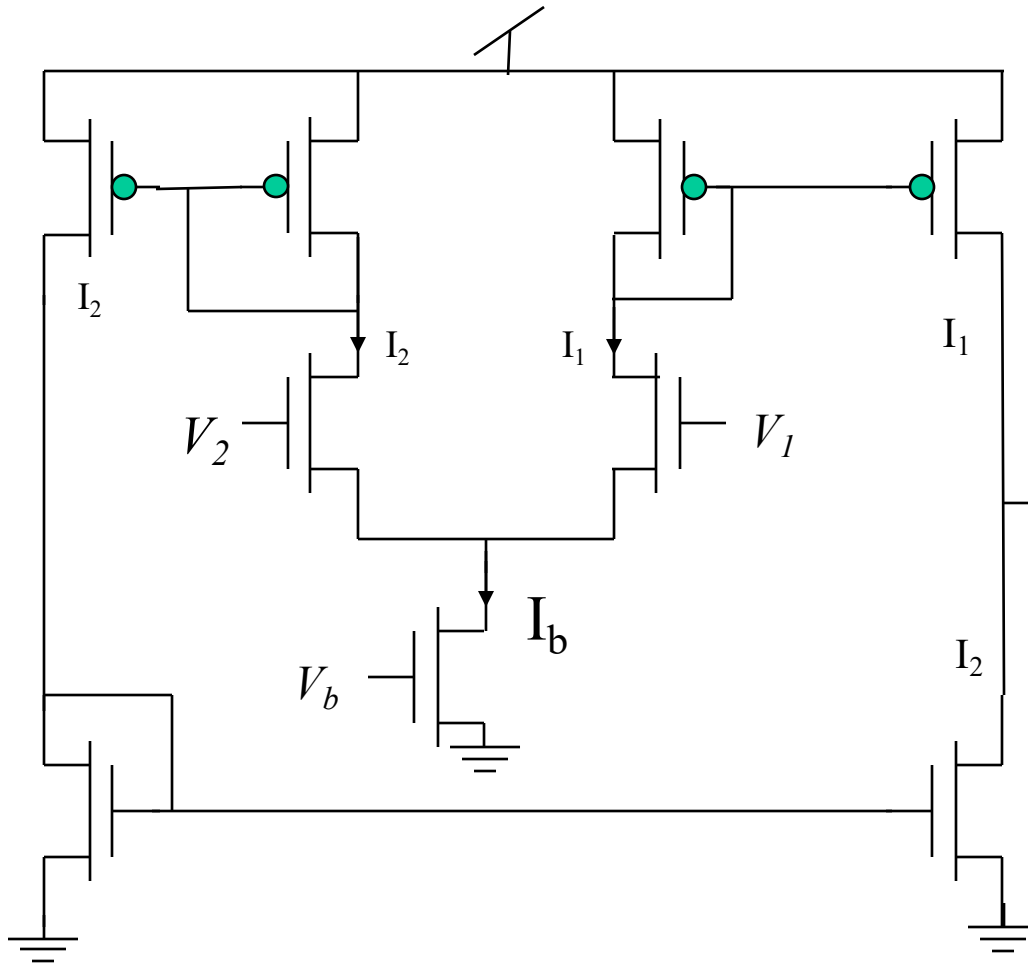


$$\begin{aligned} I_{out} &= I_1 - I_2 \\ &= I_b \tanh \left(\kappa \frac{V_1 - V_2}{2} \right) \end{aligned}$$

Good: Simple, cheap

Bad: output voltage is restricted, voltage gain is limited

Wide-range transconductance amplifier



$$I_{out} = I_b \tanh \left(\kappa \frac{V_1 - V_2}{2} \right)$$

Good: output voltage is unrestricted, gain can be $>10^4$

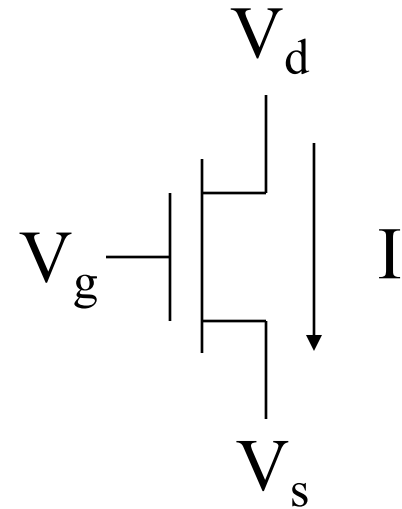
Bad: larger, more mismatch

nFET Conductances

$$I = I_0 e^{\kappa V_g / U_T} (e^{-V_s / U_T} - e^{-V_d / U_T})$$

Gate Transconductance

$$g_m = \frac{\partial I}{\partial V_g} = \frac{\kappa I}{U_T}$$



Drain Conductance

$$g_d = \frac{\partial I}{\partial V_d} = \frac{I}{V_E}; \quad V_E = \text{Early Voltage}$$

THE END

Next week: Linear systems, follower-integrator,
follower-differentiator